

# ArithmeticSite

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# Chapter 1

## The multiplicative monoid $\mathbb{N}^\times$

The Arithmetic Site, introduced by Connes and Consani, is a geometric object — a topos equipped with a structure sheaf — whose points in the topos-theoretic sense are in natural bijection with the adèle class space of  $\mathbb{Q}$ . Its construction begins with a single, elementary monoid: the positive integers under multiplication. Everything else — the tropical semiring, the presheaf topos, the structure sheaf, and ultimately the connection to the Riemann zeta function — is built on top of this one piece of data.

**Definition 1.1.** Let  $\mathbb{N}^\times$  denote the set  $\{1, 2, 3, \dots\}$  of strictly positive integers equipped with ordinary multiplication. This is a commutative monoid with unit 1. In Lean we represent this as `PNat` with its multiplicative structure, via the abbreviation `NPos`.

The unique factorisation theorem says that  $\mathbb{N}^\times$  is the free commutative monoid on the set of primes: every element factors uniquely as a finite product  $\prod_p p^{a_p}$ . This arithmetic richness — the interplay of primes, divisibility, and factorisation — is what makes  $\mathbb{N}^\times$  the right monoid for the Arithmetic Site. As we will see, it is ultimately responsible for the fact that the points of  $\widehat{\mathbb{N}^\times}$  are parametrised by the adèle class space of  $\mathbb{Q}$ , and that the Hasse–Weil zeta function of the site recovers the Riemann zeta function.

**Definition 1.2.** The *classifying category*  $B\mathbb{N}^\times$  of  $\mathbb{N}^\times$  is the one-object category whose single object is  $*$  and whose morphisms  $*$   $\rightarrow$   $*$  are the elements of  $\mathbb{N}^\times$ , with composition given by multiplication.

**Remark 1.3.** The passage from a monoid  $M$  to the one-object category  $BM$  is the standard *delooping* construction. A presheaf on  $BM$  — that is, a functor  $BM^{\text{op}} \rightarrow \mathbf{Set}$  — is exactly a set  $X$  equipped with a left action of  $M$ : each element  $n \in M$  acts as a function  $X \rightarrow X$ , and the action is compatible with composition. (Left and right actions coincide here since  $\mathbb{N}^\times$  is commutative.) The presheaf topos  $\widehat{B\mathbb{N}^\times}$  introduced in Chapter ?? is precisely the category of such sets-with- $\mathbb{N}^\times$ -action, and it is in this topos that the Arithmetic Site lives.

## Chapter 2

# The tropical semiring $\bar{\mathbb{N}}$

The tropical semiring  $\bar{\mathbb{N}}$  is the second fundamental ingredient of the Arithmetic Site. It plays the role of the *structure sheaf*: as Connes and Consani put it,  $\bar{\mathbb{N}}$  is the semiring on which  $\mathbb{N}^\times$  acts by multiplication, and it is this action that makes  $\bar{\mathbb{N}}$  into a sheaf of semirings over the topos  $\widehat{B\mathbb{N}^\times}$ . The arithmetic of  $\bar{\mathbb{N}}$  is characteristic-1 arithmetic — addition is idempotent — which is the tropical, or min-plus, shadow of ordinary arithmetic. Connes and Consani define  $\bar{\mathbb{N}}$  explicitly as  $(\mathbb{N} \cup \{+\infty\}, \inf, +)$ , confirming the min-plus convention: addition is min, with  $+\infty$  as the additive identity. When elements of  $\bar{\mathbb{N}}$  are written as exponents  $q^x$  for  $q \in (0, 1)$ , the addition becomes max on values (since smaller exponent means larger value), which is why the stalks in the paper appear in max-plus form as  $[0, 1]_{\max}$  and  $\mathbb{R}_{\max}$ .

**Definition 2.1.** The *tropical semiring*  $\bar{\mathbb{N}}$  is the set  $\mathbb{N} \cup \{+\infty\}$  equipped with:

- *Tropical addition*:  $a \oplus b := \min(a, b)$ , with identity element  $+\infty$ .
- *Tropical multiplication*:  $a \odot b := a + b$  (ordinary addition), with identity element 0.

In Lean,  $\bar{\mathbb{N}}$  is represented as `Tropical (WithTop ℕ)`, via the abbreviation `NBar`. The zero element is `trop ⊤` (i.e.  $+\infty$ ) and the multiplicative identity is `trop 0`.

**Lemma 2.2.**  $\bar{\mathbb{N}}$  is a commutative semiring.

*Proof.* This is the Mathlib instance `CommSemiring (Tropical (WithTop ℕ))`, found by `inferInstance`. □

**Lemma 2.3.** Tropical addition on  $\bar{\mathbb{N}}$  is idempotent:  $a \oplus a = a$  for all  $a \in \bar{\mathbb{N}}$ .

*Proof.*  $a \oplus a = \min(a, a) = a$ . In Lean this follows from `Tropical.add_eq_left`. □

**Remark 2.4.** Idempotency of addition — equivalently, the identity  $1 \oplus 1 = 1$  — is the defining feature of *characteristic 1* arithmetic. It is what places  $\bar{\mathbb{N}}$  in the same conceptual family as finite fields  $\mathbb{F}_p$  (characteristic  $p$ ), with min playing the role that  $+\text{ mod } p$  plays there. The scaling endomorphisms defined below are the characteristic-1 analogue of the Frobenius endomorphism  $x \mapsto x^p$  of  $\mathbb{F}_p$ .

**Definition 2.5.** For each  $n \in \mathbb{N}^\times$ , the *scaling endomorphism*  $\varphi_n : \bar{\mathbb{N}} \rightarrow \bar{\mathbb{N}}$  is defined by

$$\varphi_n(x) := nx \quad (x \in \mathbb{N}), \quad \varphi_n(+\infty) := +\infty,$$

where  $nx$  denotes ordinary multiplication in  $\mathbb{N}$ . Equivalently,  $\varphi_n(x) = x^{\odot n}$ , the  $n$ -fold tropical product of  $x$  with itself. In Lean,  $\varphi_n$  is defined as a semiring endomorphism  $\mathbb{N}\text{Bar} \rightarrow^{+*} \mathbb{N}\text{Bar}$  via `scalingEndomorphism`.

**Lemma 2.6.** *For each  $n \in \mathbb{N}^\times$ , the map  $\varphi_n$  is a semiring endomorphism of  $\bar{\mathbb{N}}$ : it preserves both operations and both identities.*

*Proof.* Preservation of  $+\infty$  and of tropical multiplication are immediate. Preservation of tropical addition uses  $n \cdot \min(a, b) = \min(na, nb)$  for  $n > 0$ , which holds because scaling by  $n$  is order-preserving on  $\mathbb{N} \cup \{+\infty\}$ . In Lean this is the `map_add'` field of `scalingEndomorphism`, proved via monotonicity of `nsmul` and a case split on `le_total`.  $\square$

**Lemma 2.7.** *The scaling endomorphisms compose multiplicatively:*

$$\varphi_m \circ \varphi_n = \varphi_{mn}$$

for all  $m, n \in \mathbb{N}^\times$ . In particular, the map  $n \mapsto \varphi_n$  is a monoid homomorphism from  $(\mathbb{N}^\times, \times)$  into the semiring endomorphism monoid  $\text{End}_{\text{Semiring}}(\bar{\mathbb{N}})$ .

*Proof.*  $(\varphi_m \circ \varphi_n)(x) = m(nx) = (mn)x = \varphi_{mn}(x)$ . In Lean this is `scalingEndomorphism_comp`, proved via `pow_mul`.  $\square$

**Remark 2.8.** The Frobenius correspondences  $\Psi(\lambda)$ , parametrised by  $\lambda \in \mathbb{Q}_+^\times$ , are a more refined object: they live in the square  $\widehat{B\mathbb{N}^{\times 2}}$  of the Arithmetic Site and are defined via the action of  $\mathbb{N}^\times \times \mathbb{N}^\times$  on  $\bar{\mathbb{N}} \otimes_{\mathbb{B}} \bar{\mathbb{N}}$ . They are introduced in Chapter ??.

## Chapter 3

# The presheaf topos $\widehat{\mathbb{N}^\times}$

The topos underlying the Arithmetic Site is the simplest kind: a presheaf topos. Its objects are sets equipped with a left  $\mathbb{N}^\times$ -action by arbitrary functions — not necessarily bijections, since  $\mathbb{N}^\times$  is a monoid rather than a group — and its morphisms are  $\mathbb{N}^\times$ -equivariant maps between them. The multiplicative structure of  $\mathbb{N}^\times$  is entirely encoded in the composition of these actions: if  $n$  and  $m$  act on a set  $X$ , then  $nm$  acts as the composite  $x \mapsto n \cdot (m \cdot x)$ .

Connes and Consani note that  $\widehat{\mathbb{N}^\times} \simeq \text{Sh}(\mathbb{N}^\times, J)$ , where  $J$  is the *chaotic topology* on  $\mathbb{N}^\times$  — the topology in which every presheaf is already a sheaf. This means there are no genuine gluing conditions to check: the topos is simply the presheaf category, with no further data. The covering families arising from factorizations of integers, foreshadowed in Chapter 1, reflect a finer site structure that one *can* put on  $\mathbb{N}^\times$ ; the Arithmetic Site itself uses the coarser chaotic topology, where that combinatorial structure is not needed.

**Definition 3.1.** The *presheaf topos*  $\widehat{\mathbb{N}^\times}$  is the category of functors  $B\mathbb{N}^{\times\text{op}} \rightarrow \mathbf{Set}$ , i.e. the category of sets equipped with a left  $\mathbb{N}^\times$ -action. In Lean this is `Action (Type*) (Multiplicative ℕ+)`.

**Lemma 3.2.**  $\widehat{\mathbb{N}^\times}$  is a Grothendieck topos.

*Proof.* Any presheaf category on a small category is a Grothendieck topos. The site is  $(\mathbb{N}^\times, J)$  where  $J$  is the chaotic topology.  $\square$

**Remark 3.3.** The points of  $\widehat{\mathbb{N}^\times}$  — geometric morphisms from  $\mathbf{Set}$  to  $\widehat{\mathbb{N}^\times}$  — are the deep link to number theory. Connes and Consani prove (Theorem 2.2 of their 2014 paper) that the category of points of  $\widehat{\mathbb{N}^\times}$  is canonically equivalent to the category of totally ordered groups isomorphic to non-trivial subgroups of  $(\mathbb{Q}, \mathbb{Q}_+)$ . The space of isomorphism classes of such points is canonically isomorphic to the double quotient

$$\mathbb{Q}_+^\times \backslash \mathbb{A}^f / \widehat{\mathbb{Z}}^*,$$

where  $\mathbb{A}^f$  is the ring of finite adèles of  $\mathbb{Q}$  and  $\widehat{\mathbb{Z}}^* = \widehat{\mathbb{Z}}^\times$  is the group of units of the profinite completion of  $\mathbb{Z}$ . This quotient is a piece of the adèle class space of  $\mathbb{Q}$ , the same space that appears in Connes’ noncommutative-geometric approach to the Riemann hypothesis. The structure sheaf  $\mathcal{O} = \widehat{\mathbb{N}}$  then equips this space with characteristic-1 geometry, whose Hasse–Weil zeta function recovers the Riemann zeta function.

## Chapter 4

# The structure sheaf

In classical algebraic geometry, a *ringed space* is a topological space equipped with a sheaf of commutative rings — the *structure sheaf* — whose stalks encode the local algebra of the space. The Arithmetic Site is a *ringed topos* in the same spirit: the role of the topological space is played by the presheaf topos  $\widehat{\mathbb{N}^\times}$ , and the role of the structure sheaf is played by  $\bar{\mathbb{N}}$ , viewed as a semiring object living inside that topos.

An object of  $\widehat{\mathbb{N}^\times}$  is a set equipped with a left  $\mathbb{N}^\times$ -action, so for  $\bar{\mathbb{N}}$  to be a semiring *object* of  $\widehat{\mathbb{N}^\times}$  the action must be by semiring endomorphisms: each scaling map  $\varphi_n$  must preserve both tropical operations and both identities. This is exactly the content of Lemma ??, which is therefore the key structural fact underlying the definition below.

**Definition 4.1.** The *structure sheaf*  $\mathcal{O}$  of the Arithmetic Site is the object of  $\widehat{\mathbb{N}^\times}$  whose underlying set is  $\bar{\mathbb{N}}$  and whose  $\mathbb{N}^\times$ -action is given by the scaling endomorphisms:  $n \cdot x := \varphi_n(x) = nx$ .

**Lemma 4.2.** The assignment  $n \mapsto \varphi_n$  defines a monoid homomorphism  $(\mathbb{N}^\times, \times) \rightarrow \text{End}_{\text{Semiring}}(\bar{\mathbb{N}})$ , making  $\bar{\mathbb{N}}$  into a semiring object of  $\widehat{\mathbb{N}^\times}$  with  $\mathbb{N}^\times$  acting by multiplication.

*Proof.* Each  $\varphi_n$  is a semiring endomorphism by Lemma ??. The homomorphism property  $\varphi_{mn} = \varphi_m \circ \varphi_n$  is Lemma ??, and  $\varphi_1 = \text{id}$  since  $\varphi_1(x) = 1 \cdot x = x$ . In Lean this is `scalingEndomorphism_comp`.  $\square$

**Remark 4.3.** Since the Arithmetic Site uses the chaotic topology, every presheaf on  $B\mathbb{N}^\times$  is automatically a sheaf — there are no gluing conditions to verify. The content of the structure sheaf is therefore entirely algebraic:  $\bar{\mathbb{N}}$  is not merely a set with  $\mathbb{N}^\times$ -action, but a semiring whose operations are *preserved* by that action. This is what Lemma ?? records.

**Remark 4.4.** Connes and Consani (Theorem 2.4 of their 2014 paper) describe the stalks of  $\mathcal{O}$  explicitly. At the point corresponding to an intermediate semifield  $F \subset K \subset \bar{F}$ , the stalk is

$$\mathcal{O}_K = \{r \in K \mid r \oplus 1 = 1\},$$

the subsemiring of elements that are “bounded above by 1” in the ordering of  $K$ . This is the characteristic-1 analogue of the ring of integers of a local field: just as  $\mathcal{O}_{K_{\mathfrak{p}}} = \{x \in K_{\mathfrak{p}} \mid |x| \leq 1\}$  in  $p$ -adic geometry, the stalk of  $\mathcal{O}$  picks out the elements of norm at most 1. At the “generic” point (corresponding to  $K = F = Z_{\max}$ ) the stalk recovers  $\bar{\mathbb{N}}$  itself.

## Chapter 5

# The Arithmetic Site

**Definition 5.1.** The *Arithmetic Site* is the pair  $(\widehat{\mathbb{N}^\times}, \mathcal{O})$  consisting of the presheaf topos  $\widehat{\mathbb{N}^\times}$  together with its structure sheaf  $\mathcal{O}$ .

## Chapter 6

# Points of the topos

**Definition 6.1.** A *point* of a topos  $\mathcal{E}$  is a geometric morphism  $p : \mathbf{Set} \rightarrow \mathcal{E}$ , i.e. an adjoint pair  $p^* \dashv p_*$  where  $p^*$  is left exact.

**Theorem 6.2.** *The points of  $\widehat{\mathbb{N}^\times}$  are in natural bijection with the adèle class space  $\mathbb{Q}_+^\times \backslash \mathbb{A}^f / \widehat{\mathbb{Z}}^*$ .*

*Proof.* (sorry)

□



## Chapter 7

# The Frobenius correspondences

**Definition 7.1.** For each  $\lambda \in \mathbb{Q}_+^\times$ , the *Frobenius correspondence*  $\Phi_\lambda$  is a correspondence on  $\widehat{\mathbb{N}^\times} \times \widehat{\mathbb{N}^\times}$  arising from the scaling action of  $\mathbb{Q}_+^\times$  on the adèle class space.

**Theorem 7.2.** *The Frobenius correspondences realize the scaling action of  $\mathbb{Q}_+^\times$  on the adèle class space at the level of the topos  $\widehat{\mathbb{N}^\times}$ .*

*Proof.* (sorry)

□

## Chapter 8

# Connection to the Riemann zeta function

**Theorem 8.1.** *The Hasse–Weil zeta function of the Arithmetic Site recovers the Riemann zeta function  $\zeta(s)$ .*

*Proof.* (sorry)

□